Review plan · for the teacher

Models and predictions: unit check

Data science · Level 3 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Lines that predict
- Training and testing
- Nearest neighbours
- Judging a model

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet for its answers.
2. Quiz (20 min): Read each question aloud; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask a team to explain it with an example from the unit.
4. Wrap-up (8 min): Work through the question most teams got wrong, together on the board.

Discussion

- Which idea will you use first?
- Which school model would you question?
- Which mistake costs most at your school?

Exit ticket

- One prediction my class could test fairly this week.

Try it at school tomorrow
Run the hidden-cards activity with a class, and ask them to judge each rule by its test error, not its training error.

Quiz sheet · for each team

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Team _____ Date _____

1 **Use the line.**

mark = 6 + 3 × hours, so 2 hours predicts _____ marks.

Word bank: 9 · 12

2 **A line fitted to 0–5 hours can be trusted at 40 hours.**

◯ True ◯ False

3 **Which score is an honest estimate for new pupils?**

◯ The test score ◯ The training score ◯ The best score ever seen

4 **Put the steps in order.**

☐ Train on the training set

☐ Test once on the test set

☐ Split the data

5 **Match each term to its meaning.**

Overfitting	• • • •	The score above which the model says yes
Classification	• • • •	Fits the noise in training data
Threshold	• • • •	Predicts a label

6 **With k = 5, the neighbours are A, B, A, A, B. Which label wins?**

◯ B ◯ A ◯ It is a tie

7 **A model scores 90% accuracy. What else should you check? Choose all that apply.**

☐ How many cases it misses ☐ Whether it was scored on test data

☐ How many false alarms it gives ☐ The colour of its chart

8 **Lowering the threshold catches more cases and raises more false alarms.**

◯ True ◯ False

9 **Training error 1, test error 9. What is most likely?**

◯ Overfitting ◯ A perfect model ◯ The test set is too easy

10 **A model's warning about a pupil should be checked by a teacher before acting.**

◯ True ◯ False

Answers · for the teacher

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1 **mark = 6 + 3 × hours, so 2 hours predicts 12 marks.**

6 + 3 × 2 = 12.

2 **False**

40 hours is far outside the data the line learned from.

3 **The test score**

The test set is data the model never saw.

4 **Top to bottom: 2, 3, 1**

Split first so the test stays unseen.

5 **Overfitting — Fits the noise in training data**
Classification — Predicts a label
Threshold — The score above which the model says yes

Three key ideas from the unit.

6 **A**

A has three votes to B's two.

7 **How many cases it misses; How many false alarms it gives; Whether it was scored on test data**

Misses, false alarms and an honest test all matter; the colour doesn't.

8 **True**

A lower bar says yes more often, rightly and wrongly.

9 **Overfitting**

A big gap between the two errors is the sign of overfitting.

10 **True**

Models make mistakes; people decide what to do.